

CLOSE WATCH

Nasdaq Closing-Auction Imbalances, Two Strategies, and a Cost Alert

Code: <https://github.com/lili1501/Gator-Quant-hacks> (python run_all.py)

1. Summary

Every day at 4:00 PM, Nasdaq sets each stock's closing price in an auction. When buy and sell orders do not match, Nasdaq publishes the gap, the **imbalance**. Using 100 large Nasdaq stocks over two years (year 1 to build; year 2 to test once), we find that **the imbalance pushes the closing price its way**: the largest year-2 imbalances produce a $+2.07\text{ bps}$ move ($t = 3.2$), predictable at 3:55 PM. However, **neither strategy survives trading costs out of sample**. An overnight fade falls from Sharpe 2.48 to 0.84 and is not significant; a pre-registered strategy trading *with* the push captures 3.6 bps but pays 5.7 bps in costs (Sharpe -1.76). The robust economic quantity is the *cost* imposed on funds forced to trade at the close: approximately \$2,000 per \$10 million order. We therefore frame the signal as a **Closing Cost Alert** estimated at 3:55 PM.

2. Economic Hypothesis

Index funds and ETFs must trade at the official close to match their benchmarks. If their orders lean one way, the auction clears only after price moves enough to attract the other side. The imbalance is a demand for liquidity, and the supplier of that liquidity should be paid. Because much of this demand is mechanical rather than news-based, the closing displacement may partly reverse overnight. This mechanism gives six testable hypotheses.

#	Hypothesis	Economic mechanism
H1	Large imbalances push prices their way from 3:55 PM to the close	The auction must clear
H2	The push partly reverses overnight	Forced flow, not information
H3	Imbalances that grew steadily and never flipped reverse more	Mechanical versus contested orders
H4	Fading large imbalances overnight is profitable after costs	Follows from H2 and H3
H5	The push can be predicted at 3:55 PM	Size, lopsidedness, and growth
H6	Trading with the imbalance at 3:55 and exiting at the close is profitable	The exit supplies missing liquidity

Pre-registration. H6, its selection rule, and its success criteria (year-2 Sharpe > 1 , $t > 2$, positive without its best day) were committed to `HYPOTHESES.md` (commit 2f7ba89) before H6 was first run. H1–H5 were explored earlier and are reported as such.

3. Data & Universe

- **Imbalance messages** (Databento `XNAS.ITCH`, `imbalance`): every Nasdaq closing-auction update from 3:45 to 4:01 PM, about one message per stock per second ($\approx 500\text{ MB}$).
- **Quotes** (`XNAS.ITCH`, `bbo-1s`): the first Nasdaq bid and ask after 15:55:00, representing the executable entry price.
- **Daily bars** (`EQUUS.SUMMARY`, `ohlcv-1d`): official open, close, and volume.

The universe comprises 100 large Nasdaq-listed stocks from October 2024 through September 2026: 496 trading days (half-days removed) and 49,560 stock-days. We exclude 1,166 rows affected by insufficient volume history, stock splits, or missing data. Year 1 (October 2024–September 2025) is

used for every rule and threshold; year 2 (October 2025–September 2026) is evaluated once per frozen rule and never used for tuning.

4. Methodology

Each stock-day uses only information available by 15:55:00. Imbalance size is dollar imbalance divided by trailing 20-day average dollar volume (ADV); additional variables are growth since the first message, direction flips, and imbalance divided by matched shares. The *push* is the 3:55 PM-to-close return; the *bounce* is the close-to-next-open return. Both are market-adjusted by subtracting the equal-weight move of all 100 stocks.

Overnight fade (H2–H4). For imbalances at least 6.69% of ADV (the year-1 95th percentile) that grew and never flipped, trade against the imbalance in the closing auction and exit in the next opening auction. Trades are equal-weighted by day; round-trip cost is 3 *bps* (6 *bps* in a stress test).

Closing Pressure (H6). At 3:55 PM, buy at the ask or sell at the bid in the imbalance direction and exit at the official close. Costs equal the observed half-spread plus 0.5 *bps* fees; residual market exposure is hedged with the 100-stock basket at 0.5 *bps*. Each stock is capped at 10% of capital. The frozen rule maximizes year-1 Sharpe over 12 declared variants: signal (raw/model) $\times$ tail (20/10/5%) $\times$ filter (on/off).

Push model and inference. The frozen linear model uses $\sqrt{\text{imbalance size}}$, lopsidedness, and growth. Random forest, gradient boosting, and XGBoost are estimated on the same three and ten inputs with fixed settings. Returns are first averaged by date; annualized Sharpe ratios and *t*-statistics are then computed from daily portfolio returns. The deflated Sharpe ratio adjusts for 32 strategy variants tested.

5. Results

The push is real (H1) and predictable (H5). Figure 1 sorts stock-days into ten groups using year-1 imbalance cutoffs. The largest buy and sell imbalances move the close in their direction in both samples.

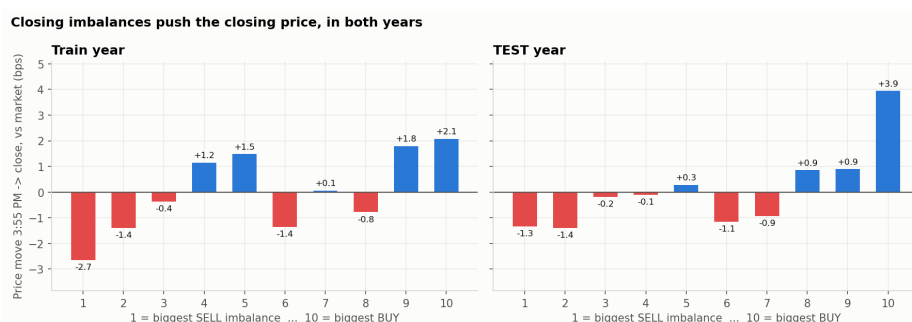


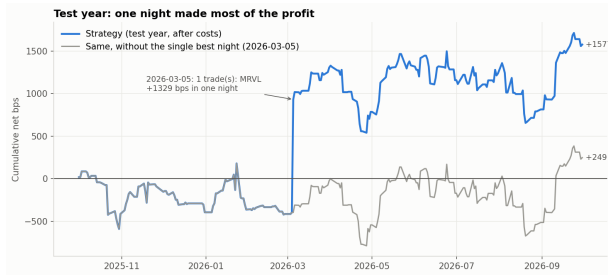
Figure 1: H1: closing-price pressure. Average market-adjusted 3:55 PM-to-close return by imbalance-size decile. Extreme groups move in the predicted direction in both years; small imbalances are noise.

Simpler is better out of sample. On identical inputs, the three-input linear model has the highest year-2 correlation (0.109). Ten-input XGBoost fits year 1 best (0.294) but performs worst in year 2 (0.085), consistent with overfitting (Appendix, Figure A1).

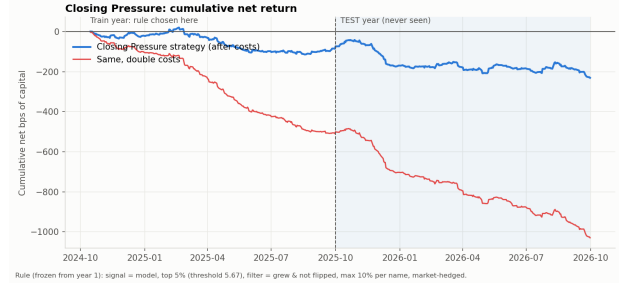
Why H6 fails—and where it does not (exploratory). Splitting Closing Pressure trades into thirds by the half-spread paid (year-1 cutoffs 1.86/4.30 *bps*) shows that spread determines the outcome. The tightest third earns +2.8 *bps* ($t = 2.5$) in year 1 and +3.6 *bps* ($t = 2.0$) in year 2; the widest third loses. Because the split was examined after H6 failed, it is a new hypothesis (H7), not a confirmed

Net of costs	Trades	bps/trade	Ann. ret.	Ann. vol.	Sharpe	t	Max DD	Turn.	DSR
<i>Overnight fade (H4), frozen rules</i>									
In-sample	769	9.75	35.1%	14.1%	2.48	2.41	-469	406×	0.67
Out-of-sample	701	4.33	16.0%	19.1%	0.84	0.83	-842	385×	0.07
<i>Closing Pressure (H6), pre-registered</i>									
In-sample	957	-0.65	-0.9%	0.7%	-1.35	-1.31	-135	202×	0.00
Out-of-sample	631	-2.03	-1.5%	0.8%	-1.76	-1.75	-189	127×	0.00

Table 1: Performance net of costs. Maximum drawdown is in bps of capital; turnover is capital traded per year. DSR is the deflated-Sharpe probability.



(a) Fade, year 2. One MRVL trade on March 5, 2026 adds 1,329 bps; without it, the year ends at 249 bps.



(b) Closing Pressure after costs (blue) and at double costs (red). It is negative already in year 1.

Figure 2: Strategy equity curves net of costs.

#	Verdict	Out-of-sample evidence (year 1 in brackets)
H1	Supported	Largest-decile push +2.07 bps, $t = 3.2$, $n = 2,109$ [+2.63, $t = 5.6$]
H2	Weak	Fade is unstable across cutoffs (top 30%: 6.1 bps; top 10%: 4.5; top 5%: 11.4)
H3	Directional	Filter raises Sharpe 0.42 $\rightarrow$ 0.84 and reduces max DD -1,405 $\rightarrow$ -842 bps
H4	Not supported	Sharpe 0.84, $t = 0.83$, DSR probability 0.07; six of twelve months positive
H5	Supported	Correlation 0.109 [0.126]; direction correct 56.9% for the largest imbalances
H6	Rejected	Captures 3.63 bps of push but pays 5.67 bps of costs; fails all preset criteria

Table 2: Hypothesis verdicts.

result.

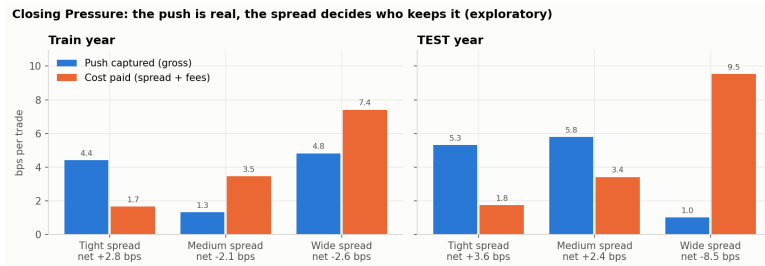


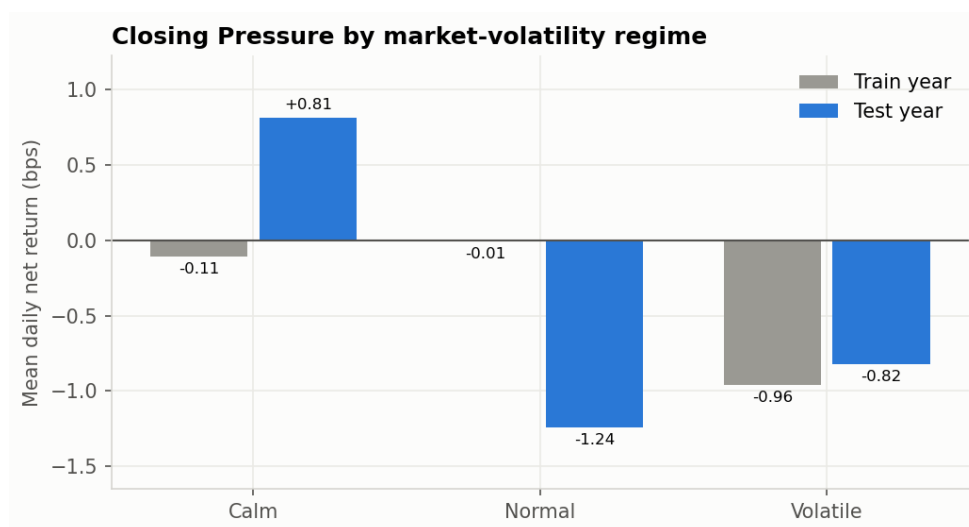
Figure 3: Closing Pressure: gross push captured versus execution cost by half-spread tercile. The signal survives only where crossing the spread is inexpensive.

6. Risk Management

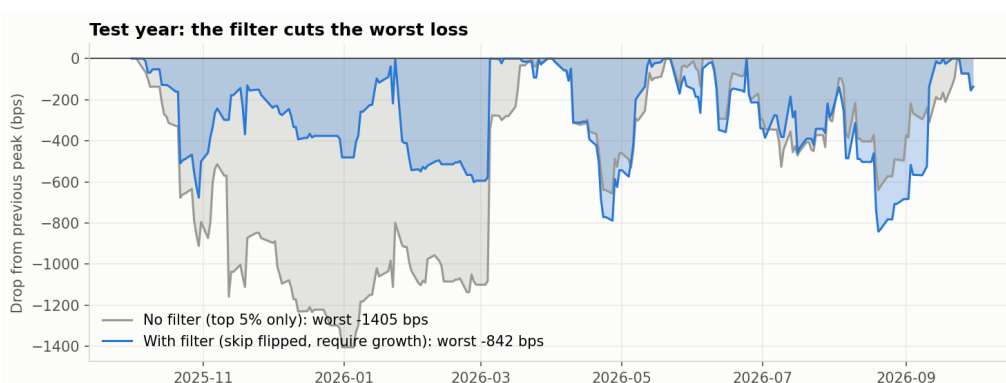
Position limits. Closing Pressure caps each stock at 10% of capital and hedges residual market exposure. The frozen fade had no cap, so one stock could constitute the entire book. A 10% cap would reduce the MRVL night from +1,329 to +133 *bps*, removing both exceptional luck and concentrated risk; every re-test should use the cap.

De-risking and hidden bets. Halve size when drawdown exceeds twice year-1 monthly volatility (39 *bps*) and restore full size after half is recovered. In year 2 this lowers Closing Pressure's drawdown from −189 to −117 *bps*, limiting loss without repairing negative expectancy. Factor regressions show that Closing Pressure partly loads on end-of-day momentum ($\beta = 0.04$, $t = 2.3$); the fade has no significant factor loadings.

Regimes. Closing Pressure earns only in calm markets and loses in normal and volatile states, including the April 2025 tariff shock. The fade remains positive across regimes, but its skewness of 5.2 reveals exposure to single-stock overnight news. Name caps and an earnings filter are therefore more important than a market hedge.



(a) Closing Pressure by market-volatility regime; tercile boundaries are fixed using year 1.



(b) Fade drawdown in year 2 with and without the pre-specified forced-flow filter.

Figure 4: Risk diagnostics. The pressure strategy is regime-dependent; the filter reduces the fade's drawdown.

7. Liquidity & Capacity

Own-price impact follows the square-root rule, $\text{cost} = k\sigma_{\text{daily}}\sqrt{\text{order}/\text{ADV}}$, with $k = 0.5$ (standard) and $k = 1.0$ (conservative). We also deduct the edge destroyed by our own opposing auction order and cap participation at 25% of the imbalance.

Year-2 annual return on capital	\$100k	\$250k	\$500k	\$1M	\$2.5M	Break-even
Overnight fade, standard	+12.2%	+10.1%	+7.6%	+4.2%	−2.7%	≈ \$1.7M
Overnight fade, conservative	+8.5%	+4.2%	−0.7%	−7.7%	−21.4%	≈ \$0.45M
Closing Pressure, standard	−1.6%	−1.8%	−2.0%	−2.4%	−3.0%	Never

Table 3: Capacity estimates. Fade profit peaks near \$41,000 per year at \$1 million. Closing Pressure’s binding cost is the spread, not modeled market impact.

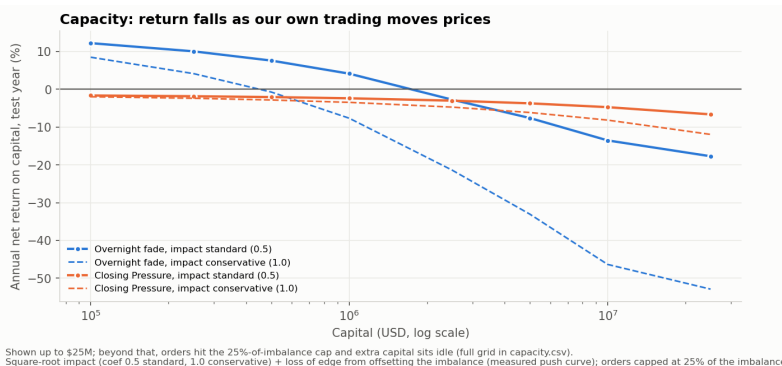


Figure 5: Annual net return versus capital deployed in year 2 under standard and conservative impact assumptions.

Economic takeaway. Both strategies are capacity constrained: a few basis points per trade, turnover of 130–400× per year, and only a handful of stocks per day. The scalable result is the cost imposed on crowded-side closing orders: approximately 2.07 *bps*, or \$2,070 per \$10 million. The Closing Cost Alert uses the 3:55 PM imbalance to estimate that cost before the auction completes.

8. Limitations & Next Steps

Limitations.

- One out-of-sample year provides limited regime coverage; using today’s 100 large Nasdaq stocks also introduces survivorship bias.
- Nasdaq BBO may overstate achievable cost relative to consolidated NBBO, while the square-root impact model is only an approximation for small orders.
- The absence of an earnings calendar leaves the fade exposed to after-hours company news.
- Broker-specific eligibility, cutoff times, and fees for late closing-auction orders require confirmation before implementation.

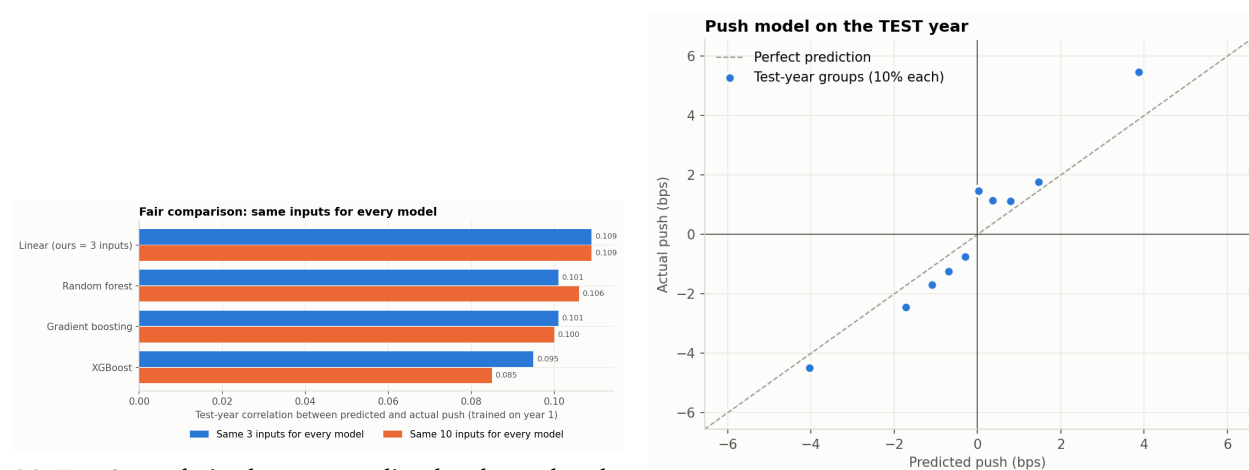
Next steps.

1. Pre-register H7—Closing Pressure only when the half-spread is below 1.86 *bps*—and test it exclusively on data after September 2026.
2. Use passive limit orders to earn rather than pay the spread.
3. Re-test the fade with a 10% name cap and an earnings-event exclusion.
4. Add consolidated quotes, sector controls, and NYSE closing auctions.

References

- Almgren, R., Thum, C., Hauptmann, E., & Li, H. (2005). Direct estimation of equity market impact. *Risk*, 18(7), 57–62.
- Bailey, D. H., & López de Prado, M. (2014). The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.
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- Databento. Nasdaq TotalView-ITCH (XNAS.ITCH) imbalance and BBO schemas; U.S. equities summary (EQUS.SUMMARY).

Appendix



A1. Year-2 correlation between predicted and actual push; all models trained on year 1.

A2. Push-model calibration in year 2 by prediction decile.

Model	Inputs	Train corr.	Test corr.	Overfit gap	Direction hit (top decile)
Linear (frozen)	3	0.126	0.109	0.017	56.9%
Linear	10	0.135	0.109	0.026	56.2%
Random forest	3/10	0.176/0.227	0.101/0.106	0.075/0.121	57.1%/56.3%
Gradient boosting	3/10	0.157/0.232	0.101/0.100	0.056/0.132	57.0%/56.4%
XGBoost	3/10	0.199/0.294	0.095/0.085	0.104/0.209	56.6%/55.6%

Table A1. Model comparison with common inputs and fixed settings.